Sreedhar Suswaram

Phoenix, AZ | (901) 283-7338 | sreedhar.sr@gmail.com

PROFESSIONAL SUMMARY

Senior engineer with 19+ years of experience in solution architecture and hands-on delivery of enterprise distributed systems, cloud-native platforms, and secure financial services. Designed payments, reconciliation, and user-management solutions at American Express, with strengths in RESTful API integration, platform modernization, reliability, and access governance. Partners with product teams, translates business requirements into technical solutions, and mentors engineers. Independent AI learning and personal proof-of-concept projects cover AI-assisted design documentation, API design, requirements analysis, and agentic systems.

TECHNICAL SKILLS

Architecture and integration: Solution architecture, distributed systems, RESTful APIs, micro-services, event-driven systems, SaaS platforms, real-time services

Backend and Data engineering: Java, Spring Boot, Vert.x, Python, FastAPI, Redis, Kafka, SQL/NoSQL

AI learning and personal POCs: Google ADK, Vertex AI Agent Engine, Gemini, OpenAI SDK, Model Context Protocol (MCP), multi-agent orchestration, RAG, vector search, guardrails

Cloud and operations: GCP, Docker, CI/CD, Splunk, Jaeger, distributed tracing, disaster recovery, production readiness; AWS and Azure familiarity

Security and governance: IAM, MFA, OAuth, RBAC, Microsoft Entra ID, SPIFFE/SPIRE, workload identity, policy-as-code, certificate lifecycle automation

PROFESSIONAL EXPERIENCE

American Express | 2013–2026

Senior Engineer, 2015–2026 | Lead Developer, 2013–2015

- Led API strategy and implementation for application-to-application services on the Digital Servicing platform and technical architecture for payment-platform modernization.
- Partnered with the portfolio architect on a cross-platform architecture proposal that resulted in a $2.35M product-funded release strategy.
- Led roadmap execution for migrating BTA-Connect clients to an in-house enterprise platform, reducing vendor dependency and improving control, scalability, and operational resilience.
- Designed and implemented enterprise-scale reconciliation, payments, and user-management platforms using Java, Vert.x, RESTful APIs, Redis, and micro-services.
- Managed up to 30 contractors across four globally distributed scrum teams; oversaw code reviews, test automation, and operational-readiness assessments.
- Built self-service user-management capabilities that reduced corporate client onboarding time by 30%.
- Built IAM and MFA frameworks for regulated payment workflows; automated certificate renewal, reducing manual effort by 90% and preventing expiration-related incidents.
- Achieved $500K in one-time cost avoidance by aligning LAC market stakeholders and architects on adopting POA instead of modifying POD.
- Partnered with UX teams to provide technical input and shape optimal designs.

EARLIER PROFESSIONAL EXPERIENCE

Verian | Technical Delivery Manager | 2010–2013

- Led architecture and delivery of e-invoicing SaaS solutions and transformed an installer-based Purchase-to-Pay product into a cloud-hosted service.
- Led e-invoicing platform development, streamlining invoice approvals and exception workflows.
- Reviewed requirements for gaps and dependencies, aligned designs with product managers, and mentored an offshore team of 10 contractors.

Harbor Payments | Engineering Roles | 2006–2010

- Developed Java/J2EE SaaS solutions for accounts payable and receivable, including EDI and XML integrations and enterprise reference-architecture alignment.
- Led a $3M migration from Mercator to Informatica PowerCenter and Business Exchange to modernize enterprise data transformation.

INDEPENDENT AI LEARNING AND PERSONAL PROJECTS

Personal learning and proof-of-concept work completed outside employment.

Travel Planner MCP

- Built an OpenAI-powered ReAct planner with planner, critic, executor, and summarizer roles, coordinating calendar, weather, and travel MCP services through a FastAPI JSON-RPC gateway.
- Implemented catalog-based tool authorization, Pydantic output contracts, bounded execution, evidence tracking, timeouts, and correlated logs using simulated domain data.

CodeGuard Code Review Assistant

- Built a Python code-review risk assessment pipeline combining Git diff analysis, Tree-sitter structure metrics, secret detection, and OSV dependency vulnerability lookups.
- Integrated OpenAI and Claude triage with JSON Schema validation and retries to produce risk assessments, evidence references, and Markdown reports from extracted metadata.

MCP Studio

- Built FastAPI-hosted MCP clients and servers to explore Streamable HTTP sessions, tool discovery, invocation, and request inspection.
- Implemented MCP progress notifications and REST Server-Sent Events using shared streaming logic; configured Docker Compose and Nginx with buffering disabled.

ADK Calendar Assistant

- Built a read-only calendar assistant using Python, Google ADK, Gemini, and FastAPI, with Google Calendar API tools for upcoming events and date-based availability queries.
- Implemented Microsoft Entra ID sign-in, delegated Google OAuth, token refresh, ADK conversation sessions, and browser chat that resumes pending questions after authentication.

EDUCATION AND CERTIFICATIONS

M.S., Electrical & Computer Engineering — University of Memphis

Harvard Certificate in Leadership Excellence